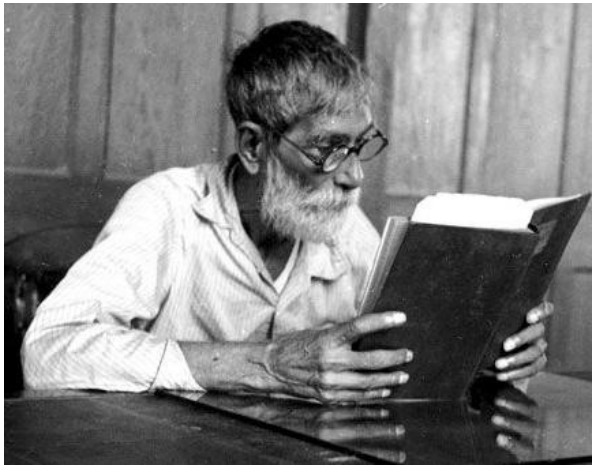
CY4001: Industrial Chemistry

Unit 1: Introduction to Chemical Industry

Prafulla Chandra Ray (P. C. Ray)

Father of Indian Chemical Industry

- Renowned chemist, academician, and industrialist
- Pioneer of modern chemical research in India
- Discovered mercurous nitrite (1896)
- Founder of Bengal Chemical & Pharmaceutical Works Ltd., India's first pharma company
- Promoted Swadeshi industry, competing with British imports
- Dedicated his entire teaching income (~15 years) to education and science
- Used science for social upliftment and disaster relief in Bengal



Ray founded Bengal Chemical and Pharmaceutical Works Limited (BCPW), the country's first drugmaker, in the early 1890s.

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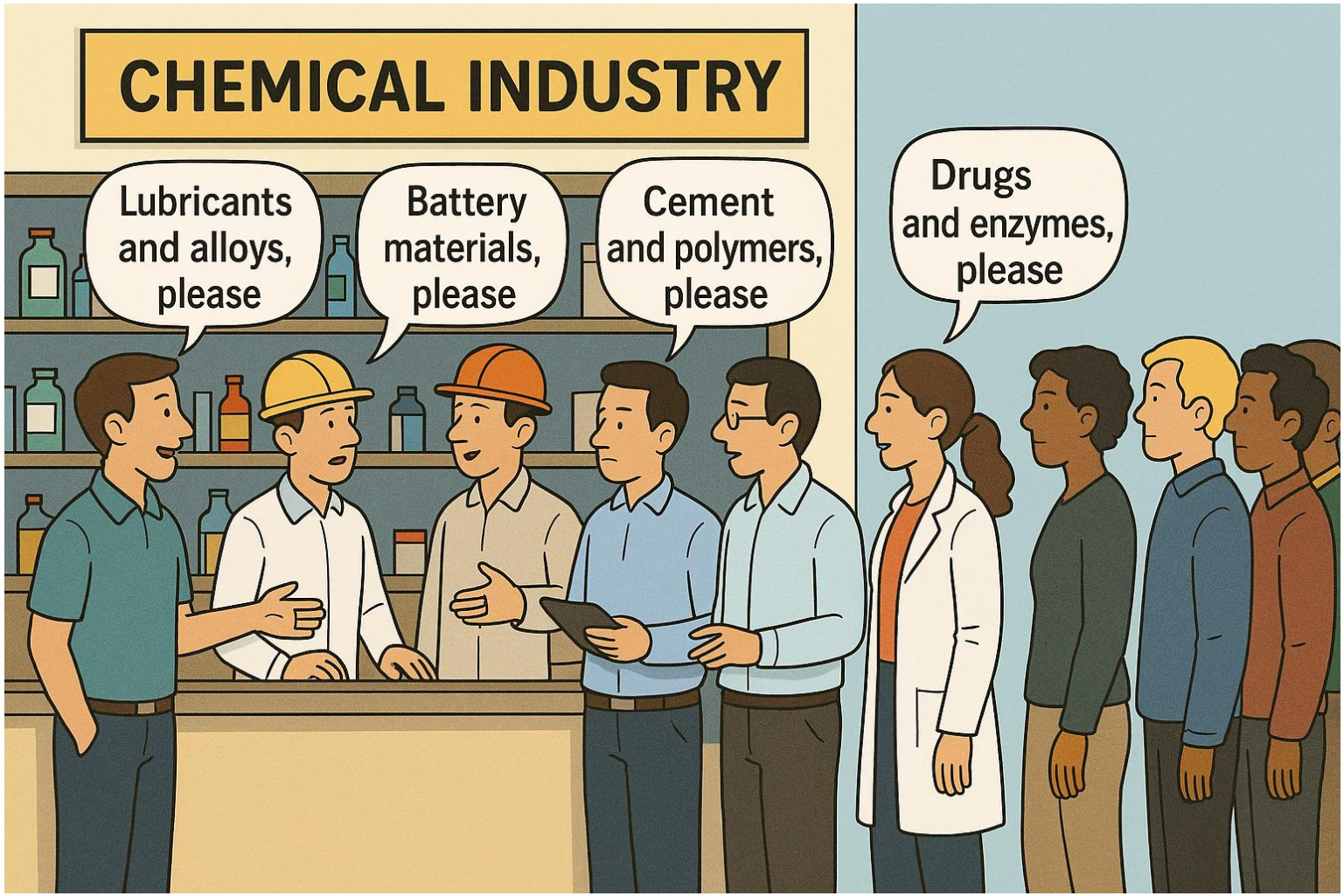

"From fuels to drugs to chips - The chemical industry powers every thing."

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Chemical Industry

- One of the oldest industries and playing an important role in the social, cultural and economic growth of a nation and in providing basic needs of humankind - food, shelter and clothing have become an indispensable part of our life.
- It is one of most diversified of all industrial sectors covering thousands of products.

History of Chemical Industry

- The birth of the heavy chemical industry (production of chemicals in large quantities for a variety of uses) coincided with the beginnings of the Industrial Revolution
- One of the first chemicals to be produced in large amounts through industrial processes was sulfuric acid. (1790s)

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Chemical Industry

Raw Materials → **Products**

- ✓ Coal
- ✓ natural gas
- ✓ Petroleum
- ✓ Air
- ✓ Water
- ✓ Salt
- ✓ Limestone
- ✓ Sulfur

- ✓ Basic chemicals: chlorine, sulfuric acid, and sodium carbonate
- ✓ Specialty chemicals: high-value chemicals used in adhesives, sealants, and paints
- ✓ Consumer chemicals: soaps and detergents
- ✓ Agricultural chemicals: fertilizers, herbicides, and pesticides
- ✓ Pharmaceuticals: Chemicals used in Antibiotics, vaccines

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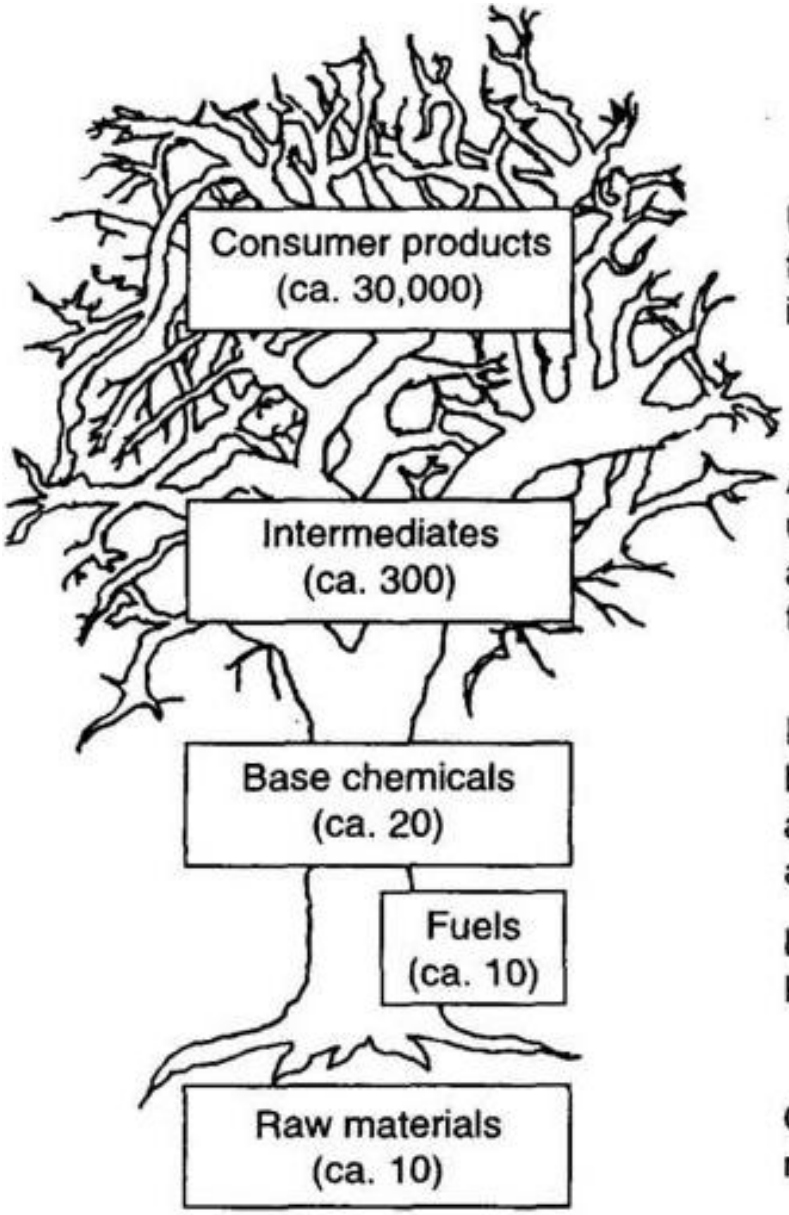
Chemical Industry Tree



Plastics, electronic materials, fibers, solvents, detergents, insecticides, pharmaceuticals — Specialty chemicals

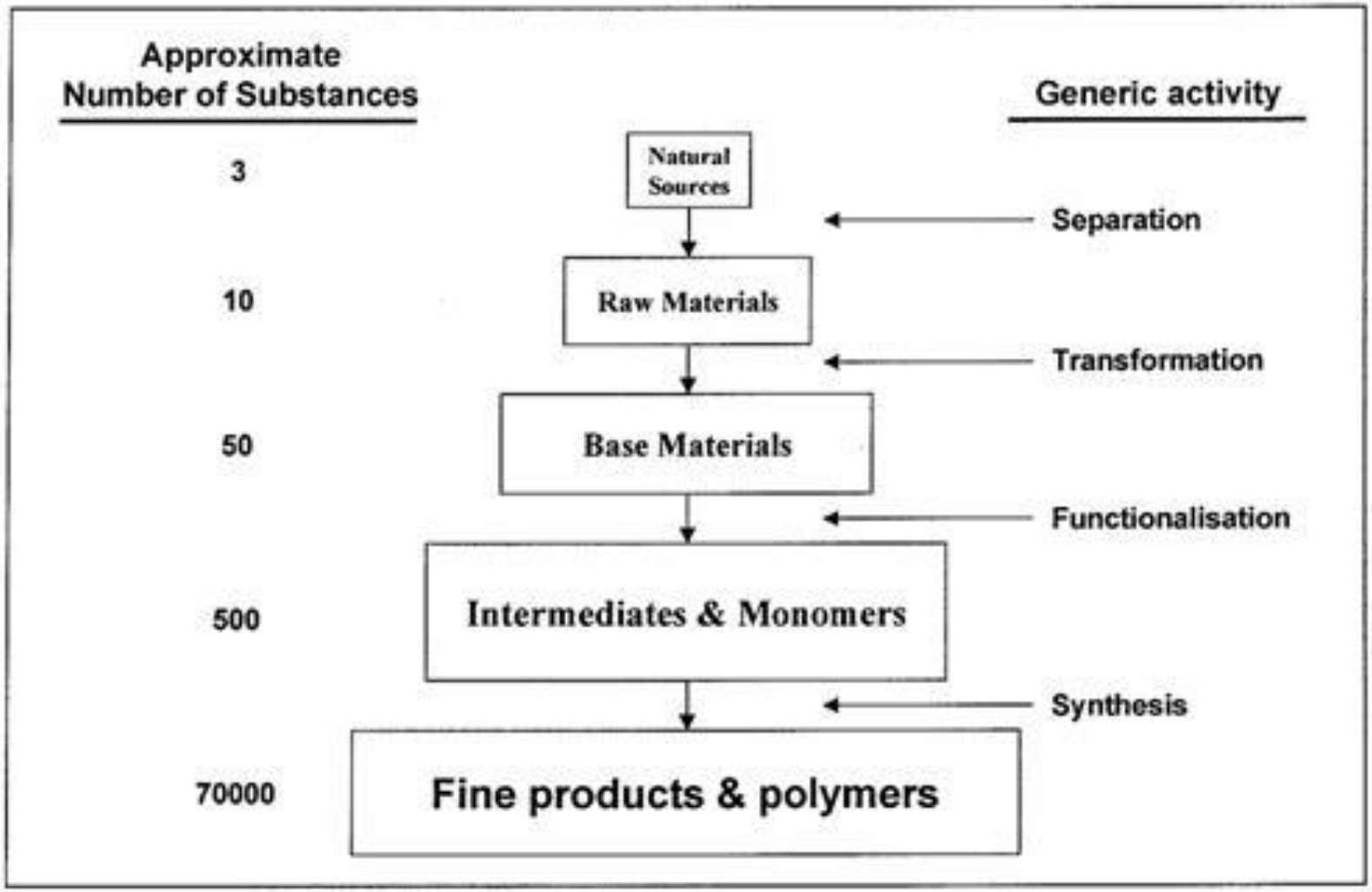
Acetic acid, formaldehyde, urea, ethene oxide, acrylonitrile, acetaldehyde, terephthalic acid

Ethene, propene, butene, benzene, synthesis gas, ammonia, methanol, sulfuric acid, chlorine

Bulk chemicals

LPG, gasoline, diesel, kerosene

Oil, natural gas, coal, biomass, rock, salt, sulfur, air, water


Approximate Number of Substances
Generic activity
3
Natural Sources
Separation
10
Raw Materials
Transformation
50
Base Materials
Functionalisation
500
Intermediates & Monomers
Synthesis
70000
Fine products & polymers

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Crude Oil, Natural gas → **Petroleum Industry** → Synthesis gas, Fuels, Mercaptanes, LPG

Naptha, Vinyl Chloride, Styrene → **Polymer Industry** → Polyethylene, Polypropene, Styrene,

Ores, Mineral salts → **Inorganic Industry** → Metals, Nitric acid, Sulphuric acids

Natural Feedstocks, Air, Organics. → **Agro Chemicals** → Pesticides, fungicides, herbicides, artificial fertilisers

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Fine chemical intermediates → **Pharma Chemicals** → Pharma proteins, synthetic drugs, antibiotics

Sugars, biomass, plant wastes → **Bio-Chemicals** → Organic acids, solvents, enzymes

Ores, Mineral salts → **Power Chemicals** → Metals, Nitric acid, Sulphuric acids

Natural products, Citric acid, other fine chemicals → **Food chemicals** → Food colors, preservatives, flavor enhancers (MSG), artificial sweenters

Introduction to Industrial Chemistry

Development of the chemical Industry

Middle Age	Oldest traces of a chemical industry primarily based on the knowledge and skill in producing candles, soaps, paints, and medicaments
1700s	Chemical production came of age as an industry but it remained small because many of the manufacturers did not have the capabilities for continual and larger production
1860s	Discovery of the first synthetic dyestuff, “mauve” by W. H. Perkins (instrumental in the evolution of the organic chemical industry)
1900s	Increasing scientific knowledge revolutionized the chemical industry
1940	Organic chemical industry has grown at a remarkable rate because of the development in petrochemical sectors
1960s and 1970s	Enormous increase in demand for synthetic polymers lead to the development of polymer based industries
After 1970s	Many of the basic processes for producing key intermediate chemicals have lost their patent protection over the years, enabling other countries of the world, to buy their own manufacturing plants

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The role of chemistry in industry

Food

Fertilizer & Agrochemical

Clothing

Synthetic fibers, Dyestuffs, Textiles, Auxiliaries, Specialty Chemicals

Shelter

Polymer composites, Coating, New Performance Materials

Health Care

Pharmaceuticals, Polymers, Synthetics, Detergent

Quality of Life

Transportation, Education, Fuel, Electricity, Energy, Water supply, Management, Communication, Polymers & Industrial Chemicals

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India is an attractive hub for chemical companies

- Chemical industry is one of the most significant contributors to the growth of any economy
- The Indian chemical industry is one of the largest worldwide. In 2020, it ranked fifth in the world in terms of revenue
- India has over 80,000 chemical products
- Chemicals constitute ~5.4% of India's total exports
- Chemical industry revenue has been growing at an average rate of 15% in the last 5 years.

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The role of chemistry in industry

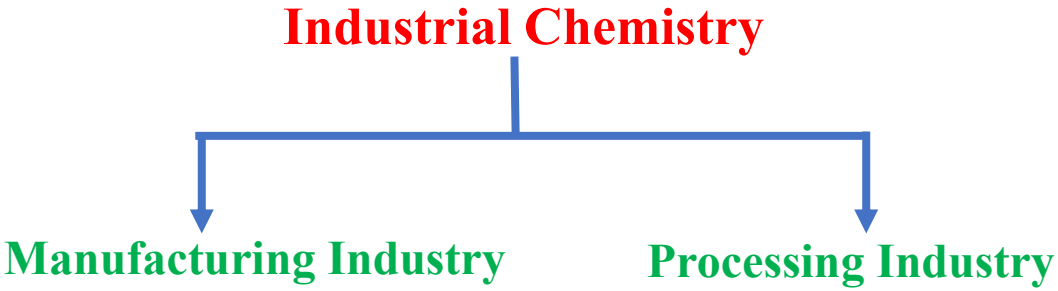
Industrial chemistry:

The branch of chemistry that deals with converting raw materials into chemical products using various chemical processes and methods.

The scope of industrial chemistry:

- Industrial chemistry is designed to develop formulation scientists for the manufacturing industry
- It is an essential as it compels industries to manufacture synthetic alternatives to protect excessive use of natural components and provide alternative options to create equilibrium.

Types of industrial chemistry:



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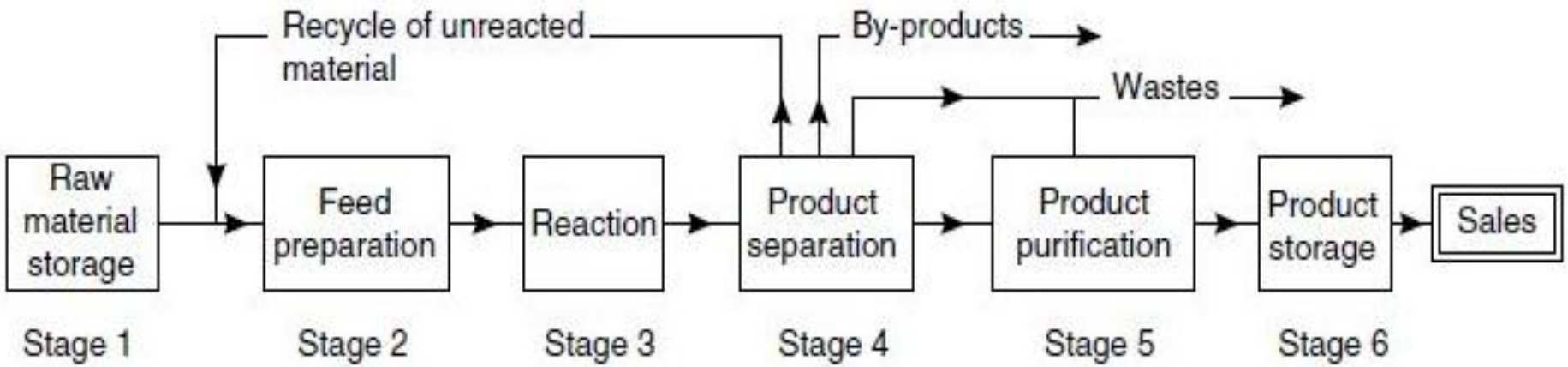
Manufacturing Industry

- Manufacturing industry deals with the manufacturing of finished new products from raw materials or essential parts.
- This process of transforming raw materials or unfinished products into new finished products can involve physical, chemical, or mechanical transformations.

Chemistry in manufacturing industry

- The transformation of organic and inorganic raw materials by a chemical process and the formulation of products.

Anatomy of a chemical industry



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Processing Industry

Processing Industry is an industry, in which the raw materials undergo various physical and chemical reaction during their processing into finished products.

Chemistry in Processing Industry (CPI)

- In CPI, the products differ chemically from the raw materials as a result of undergoing one or more chemical reactions during the manufacturing process.
- The CPI includes those manufacturing facilities whose products result from:
  - Chemical reactions between organic materials, or inorganic materials, or both.
  - Extraction, separation, purification of a natural product, with or without the aid of chemical reactions.

Chemical processes: basic chemical reactions (processes) as polymerization, oxidation, reduction, hydrogenation,

Examples of products: Plastics, resins, dyes, pharmaceuticals, paints, soaps, detergents, petrochemicals and perfumes

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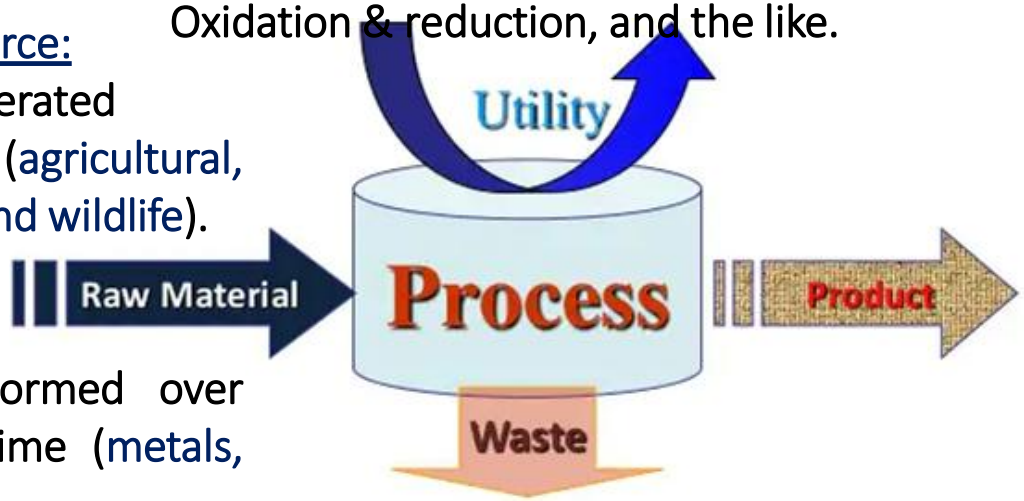
Overview of The Chemical Process Industry

Processes-basic chemical reactions:
Polymerization, hydrogenation
Oxidation & reduction, and the like.

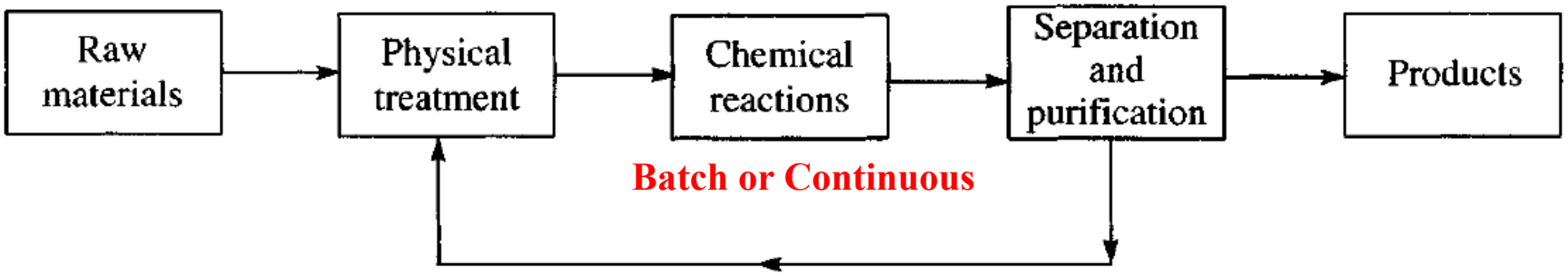
Raw-Materials Resource:

- Renewable-regenerated themselves (agricultural, forestry, fishery and wildlife).
- Non-Renewable-Formed over long period of time (metals, minerals)



- ✓ Plastics
- ✓ Resins
- ✓ Dyes
- ✓ Pharmaceuticals
- ✓ Paints
- ✓ Soaps
- ✓ detergents
- ✓ Petrochemicals
- ✓ Perfumes
- ✓ Synthetic materials

Typical Structure of CPI



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Contribution of Chemical Industry

- **The three largest sectors within the world of CPI are petrochemicals (30%), Pharmaceuticals (16.5%), and performance (speciality) chemicals (16%).**

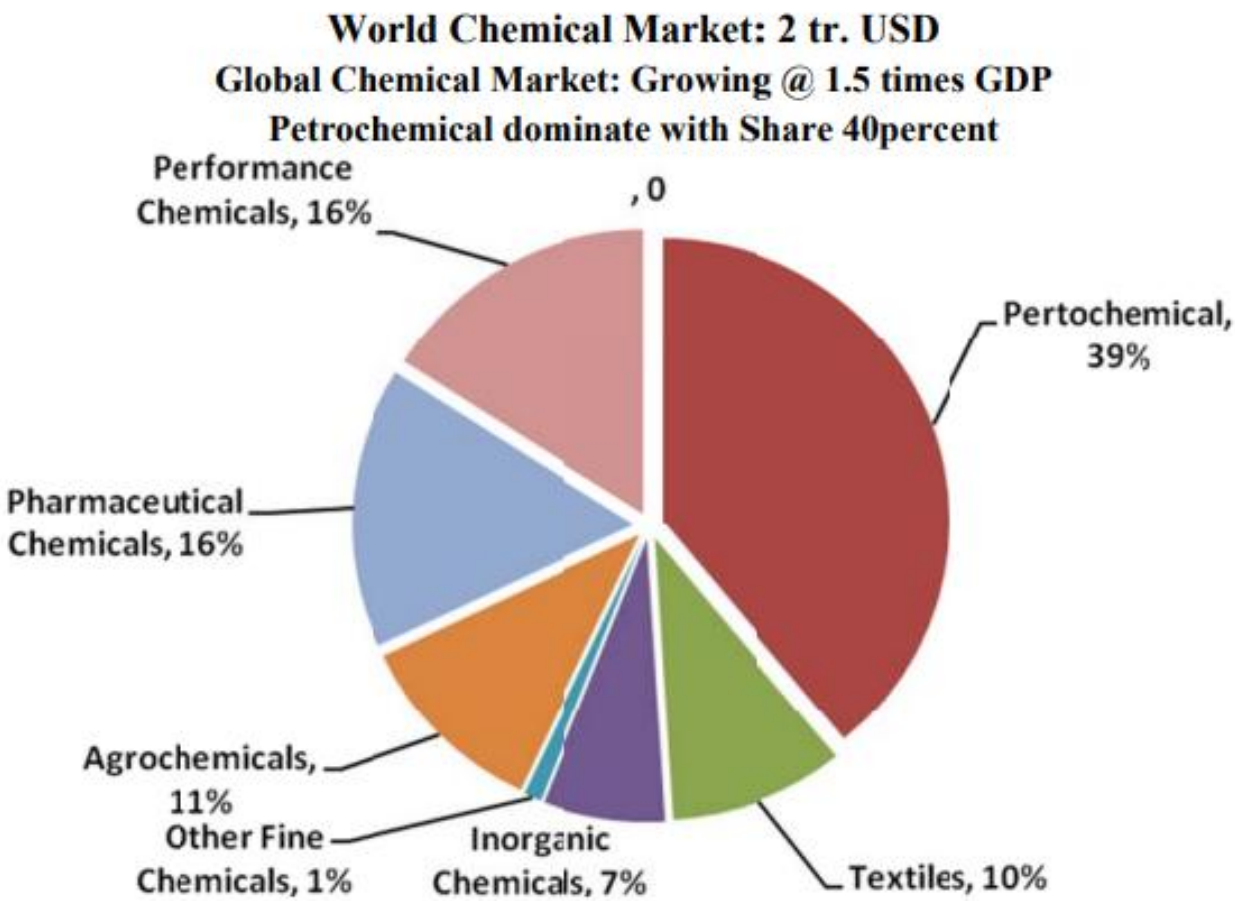

Figure M-I 1.3: World Chemical Market

Sources: Nanvaty, 2008

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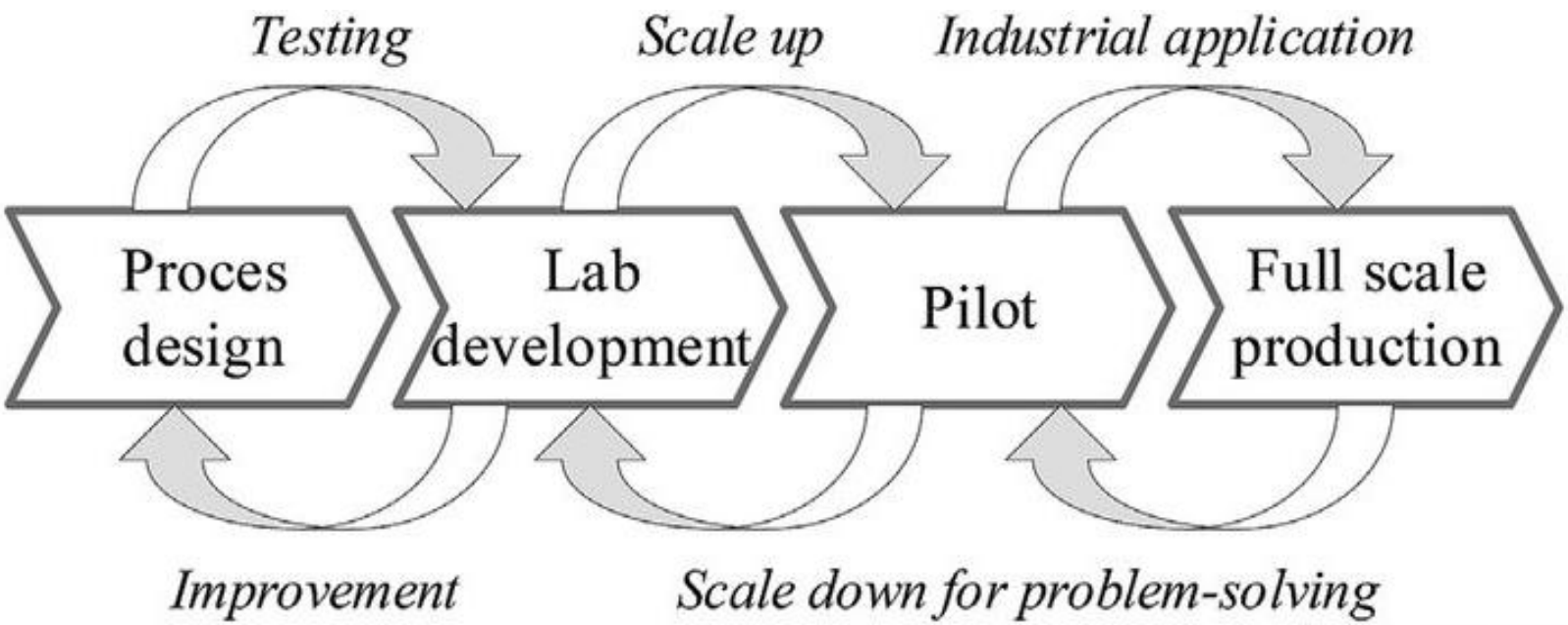
Role of chemistry in industry in scale up procedure

The scale-up process involves a range of activities that help transition a chemical process from the lab to the plant



Investigating potential process hazards; Understanding reaction kinetics and thermodynamics; Identifying and characterizing impurities; Mixing and mass transfer studies; Heat transfer and removal studies

Factors involve in moving from bench to manufacturing plant

- **Knowledge transfer:** The engineering development team must transfer knowledge and techniques to the production team. Effective training can help ensure complete knowledge transfer and reduce the time spent on faulty products

- **Safety measures:** Adequate safety measures must be developed while maintaining the economic feasibility of the process.

- **Materials handling:** Materials handling, solid-liquid separations, stream recycling, and waste streams need to be addressed in the pilot plant before the demonstration and commercial plants are built.

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Typical Issues in Chemical Processing Industries

Reduce Emissions

Save Water

Reduce Waste

Reduce Operating Costs

Minimize Wastewater

Improve Efficiency

Increase Throughout

Introduce New Plant

Meet New Product Specifications

Improve Utility System Performance

Reduce and Plan Capital

Safety Considerations in Process Industries

- The misuse or the mishandling of a simple instrument such as a knife, hammer, or sickle may result in an injury to the holder.

- Workers in a factory, a manufacturing plant, or a chemical plant remain exposed to moving conveyers, machines, dangerous chemicals, heat, pressures, high electric fields, accelerating objects, and other sources of hazards.

- Workers → not protected → injuries to death → surrounding environment → loss of life